

Exam 1 - Part 2: Statistical Analysis

AGST 50104 - Experimental Design

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Question 1

The following data refer to an experiment comparing varieties with respect to cotton properties used in fiber manufacturing. The characteristic selected for analysis was the Pressley strength index. This index was determined by measuring the breaking tension of a bundle of fibers with a pre-fixed cross-sectional area. The experiment was conducted using a randomized complete block design with three replications. The replications consisted of three strength-testing machines, each operated by a different technician. What can you conclude after proceeding with the proper analysis for this experiment? Use figures to illustrate your conclusions.

Cultivar	Machine I	Machine II	Machine III	Mean
V1	7.62	8.00	7.93	7.85
V2	8.14	8.15	7.87	8.05
V3	7.76	7.73	7.74	7.74
V4	7.17	7.57	7.80	7.51
V5	7.46	7.68	7.21	7.45
V6	7.30	7.70	7.50	7.50

Answer

Model (RCBD):

$$y_{ij} = \mu + \tau_i + \beta_j + \varepsilon_{ij}, \quad \varepsilon_{ij} \sim N(0, \sigma^2)$$

where τ_i is cultivar effect and β_j is machine (block) effect.

```
q1_data <- tibble::tribble(  
  ~cultivar, ~machine, ~strength,  
  "V1", "I", 7.62,  
  "V1", "II", 8.00,  
  "V1", "III", 7.93,  
  "V2", "I", 8.14,  
  "V2", "II", 8.15,
```

```

"V2", "III", 7.87,
"V3", "I", 7.76,
"V3", "II", 7.73,
"V3", "III", 7.74,
"V4", "I", 7.17,
"V4", "II", 7.57,
"V4", "III", 7.80,
"V5", "I", 7.46,
"V5", "II", 7.68,
"V5", "III", 7.21,
"V6", "I", 7.30,
"V6", "II", 7.70,
"V6", "III", 7.50
) |>
  mutate(cultivar = factor(cultivar), machine = factor(machine))

q1_model <- lm(strength ~ machine + cultivar, data = q1_data)

```

Assumptions

```

q1_shapiro <- shapiro.test(residuals(q1_model))
q1_levene <- leveneTest(strength ~ cultivar, data = q1_data)

q1_shapiro

```

```

#>
#> Shapiro-Wilk normality test
#>
#> data:  residuals(q1_model)
#> W = 0.97585, p-value = 0.8975

```

```
q1_levene
```

```

#> Levene's Test for Homogeneity of Variance (center = median)
#>      Df F value Pr(>F)
#> group  5  0.6615 0.6595
#>      12

```

Interpretation:

- Normality is acceptable (Shapiro-Wilk $p = 0.8975$).
- Homogeneity of variance is acceptable (Levene $p = 0.659$).

ANOVA and mean comparison

```
q1_anova <- Anova(q1_model, type = 3)
q1_anova
```

```
#> Anova Table (Type III tests)
#>
#> Response: strength
#>
#>          Sum Sq Df    F value    Pr(>F)
#> (Intercept) 134.792  1 3672.8093 3.637e-14 ***
#> machine      0.160  2   2.1744  0.16441
#> cultivar     0.856  5   4.6629  0.01858 *
#> Residuals    0.367 10
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
q1_emm <- emmeans(q1_model, ~cultivar)
q1_emm
```

```
#>   cultivar emmean    SE df lower.CL upper.CL
#> V1         7.85 0.111 10     7.60     8.10
#> V2         8.05 0.111 10     7.81     8.30
#> V3         7.74 0.111 10     7.50     7.99
#> V4         7.51 0.111 10     7.27     7.76
#> V5         7.45 0.111 10     7.20     7.70
#> V6         7.50 0.111 10     7.25     7.75
#>
#> Results are averaged over the levels of: machine
#> Confidence level used: 0.95
```

```
pairs(q1_emm, adjust = "tukey")
```

```
#>   contrast estimate    SE df t.ratio p.value
#> V1 - V2    -0.2033 0.156 10   -1.300  0.7792
#> V1 - V3     0.1067 0.156 10    0.682  0.9801
#> V1 - V4     0.3367 0.156 10    2.152  0.3355
#> V1 - V5     0.4000 0.156 10    2.557  0.1949
#> V1 - V6     0.3500 0.156 10    2.238  0.3008
#> V2 - V3     0.3100 0.156 10    1.982  0.4129
#> V2 - V4     0.5400 0.156 10    3.452  0.0516
#> V2 - V5     0.6033 0.156 10    3.857  0.0279
#> V2 - V6     0.5533 0.156 10    3.538  0.0453
#> V3 - V4     0.2300 0.156 10    1.470  0.6885
#> V3 - V5     0.2933 0.156 10    1.875  0.4662
```

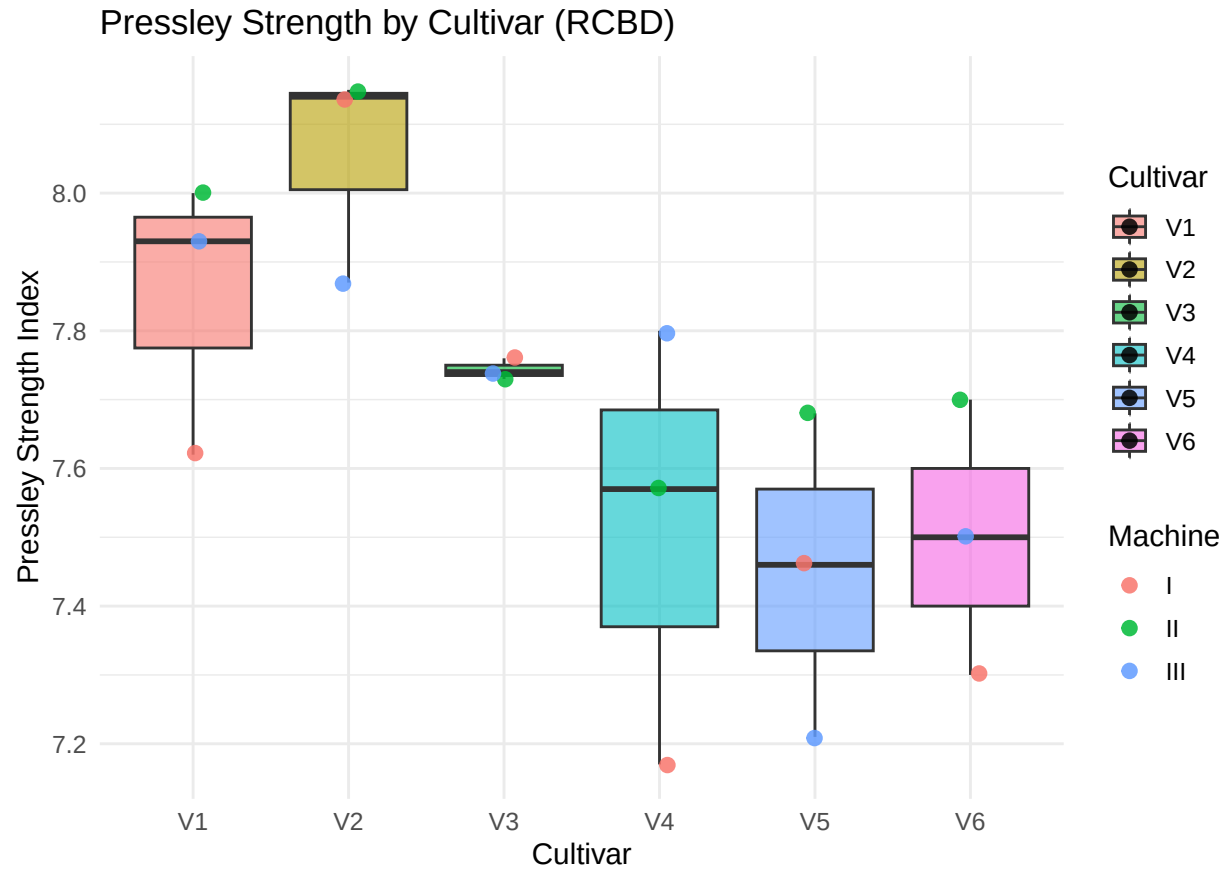
```
#> V3 - V6      0.2433 0.156 10    1.556  0.6410
#> V4 - V5      0.0633 0.156 10    0.405  0.9981
#> V4 - V6      0.0133 0.156 10    0.085  1.0000
#> V5 - V6     -0.0500 0.156 10   -0.320  0.9994
#>
#> Results are averaged over the levels of: machine
#> P value adjustment: tukey method for comparing a family of 6 estimates
```

Interpretation:

- Machine effect is not significant ($p = 0.164$).
- Cultivar effect is significant ($p = 0.0186$).
- V2 has the highest adjusted mean (about 8.05), and Tukey results show V2 is significantly higher than V5 and V6 at $\alpha = 0.05$.

Figure

```
ggplot(q1_data, aes(x = cultivar, y = strength, fill = cultivar)) +
  geom_boxplot(alpha = 0.6, outlier.shape = NA) +
  geom_point(
    aes(color = machine),
    position = position_jitter(width = 0.08),
    size = 2.4,
    alpha = 0.85
  ) +
  labs(
    title = "Pressley Strength by Cultivar (RCBD)",
    x = "Cultivar",
    y = "Pressley Strength Index",
    color = "Machine",
    fill = "Cultivar"
  ) +
  theme_minimal(base_size = 12)
```



Final conclusion for Question 1:

The RCBD analysis indicates cultivar differences are present. V2 is the best performer, while machine-to-machine variability was not statistically important in this experiment.

Question 2

The data set `bacteria.csv` is from an experiment that evaluated bacterial growth on 24 treatments. It is a Completely Randomized Design (CRD) with four replications.

- Formulate the statistical hypotheses H_0 and H_1 related to the treatments.
- Using appropriate tests, verify the basic assumptions of the statistical model for performing ANOVA. Interpret the results.
- In the event of violation of any assumptions, assess the possibility of determining an appropriate transformation using Box-Cox (1946). Then, recheck assumptions.
- Perform ANOVA on the appropriate scale at 5% significance. Present and interpret results. Identify treatment with less bacterial growth.

Answer

(a) Hypotheses

Let μ_i be the mean bacterial growth for treatment i , $i = 1, \dots, 24$.

$$H_0 : \mu_1 = \mu_2 = \dots = \mu_{24}$$

$$H_1 : \text{at least one treatment mean differs}$$

Data preparation

```
q2_data <- read.csv("bacteria.csv") |>
  pivot_longer(cols = starts_with("r"), names_to = "rep", values_to = "growth") |>
  mutate(treat = factor(treat), rep = factor(rep))

head(q2_data)
```

```
#> # A tibble: 6 x 3
#>   treat rep   growth
#>   <fct> <fct>   <int>
#> 1 1     r1       255
#> 2 1     r2       229
#> 3 1     r3       184
#> 4 1     r4       206
#> 5 2     r1       116
#> 6 2     r2        95
```

(b) Assumptions on raw scale

```
q2_model_raw <- lm(growth ~ treat, data = q2_data)

q2_raw_shapiro <- shapiro.test(residuals(q2_model_raw))
q2_raw_levene <- leveneTest(growth ~ treat, data = q2_data)

q2_raw_shapiro
```

```
#>
#> Shapiro-Wilk normality test
#>
#> data: residuals(q2_model_raw)
#> W = 0.93434, p-value = 0.0001232
```

```
q2_raw_levene
```

```
#> Levene's Test for Homogeneity of Variance (center = median)
#>      Df F value    Pr(>F)
#> group 23  2.8689 0.0003527 ***
#>      72
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Interpretation:

- Normality is violated on raw scale (Shapiro-Wilk $p = 0.000123$).
- Homogeneity is also violated on raw scale (Levene $p = 0.000353$).

(c) Box-Cox transformation and re-check

```
q2_boxcox <- boxcox(q2_model_raw, plotit = FALSE)
q2_lambda <- q2_boxcox$x[which.max(q2_boxcox$y)]
q2_lambda
```

```
#> [1] 0.2
```

```
q2_data <- q2_data |>
  mutate(growth_bc = (growth^q2_lambda - 1) / q2_lambda)

q2_model_bc <- lm(growth_bc ~ treat, data = q2_data)
```

```
q2_bc_shapiro <- shapiro.test(residuals(q2_model_bc))
q2_bc_levene <- leveneTest(growth_bc ~ treat, data = q2_data)

q2_bc_shapiro
```

```
#>
#> Shapiro-Wilk normality test
#>
#> data: residuals(q2_model_bc)
#> W = 0.98586, p-value = 0.3948
```

```
q2_bc_levene
```

```
#> Levene's Test for Homogeneity of Variance (center = median)
#>      Df F value Pr(>F)
#> group 23  1.5423 0.08462 .
#>      72
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Interpretation:

- Box-Cox suggests $\lambda \approx 0.2$.
- After transformation, assumptions are acceptable:
 - Shapiro-Wilk $p = 0.395$
 - Levene $p = 0.0846$

(d) ANOVA on appropriate scale

```
q2_anova <- anova(q2_model_bc)
broom::tidy(q2_anova)
```

```
#> # A tibble: 2 x 6
#>   term      df sumsq meansq statistic  p.value
#>   <chr>    <int> <dbl>  <dbl>    <dbl>    <dbl>
#> 1 treat      23  391.   17.0      77.6 8.62e-42
#> 2 Residuals   72  15.8   0.219     NA      NA
```

Interpretation:

- Treatment effect is highly significant ($p < 0.001$; approximately 8.62×10^{-42}).
- Reject H_0 . Not all treatment means are equal.

Treatment with less growth

```
q2_means <- q2_data |>
  group_by(treat) |>
  summarize(
    mean_raw = mean(growth),
    mean_bc = mean(growth_bc),
    .groups = "drop"
  ) |>
  arrange(mean_raw)
```

q2_means

```
#> # A tibble: 24 x 3
#>   treat mean_raw mean_bc
#>   <fct>   <dbl>   <dbl>
#> 1 20      12.8     3.31
#> 2 16      15.2     3.58
#> 3 12      20.2     4.10
#> 4 8       21.2     4.17
#> 5 4       23.8     4.29
#> 6 24      33.2     5.05
#> 7 19      34      5.08
#> 8 3       34.2     5.06
#> 9 7       36.8     5.28
#> 10 15     39      5.40
#> # i 14 more rows
```

Interpretation:

- The treatment with less bacterial growth is treatment 20 (raw mean = 12.75).

```
q2_emm <- emmeans(q2_model_bc, ~treat)
q2_emm
```

```
#>   treat emmean    SE df lower.CL upper.CL
#> 1      9.67 0.234 72    9.20    10.13
#> 2      7.56 0.234 72    7.09     8.02
#> 3      5.06 0.234 72    4.60     5.53
#> 4      4.29 0.234 72    3.83     4.76
#> 5      8.84 0.234 72    8.37     9.30
#> 6      7.56 0.234 72    7.10     8.03
#> 7      5.28 0.234 72    4.81     5.74
#> 8      4.17 0.234 72    3.70     4.64
#> 9      9.36 0.234 72    8.89     9.83
```

```

#> 10      7.55 0.234 72      7.08      8.01
#> 11      5.82 0.234 72      5.36      6.29
#> 12      4.10 0.234 72      3.63      4.56
#> 13      9.00 0.234 72      8.53      9.46
#> 14      8.25 0.234 72      7.78      8.72
#> 15      5.40 0.234 72      4.93      5.86
#> 16      3.58 0.234 72      3.11      4.05
#> 17      9.08 0.234 72      8.62      9.55
#> 18      7.92 0.234 72      7.45      8.39
#> 19      5.08 0.234 72      4.62      5.55
#> 20      3.31 0.234 72      2.85      3.78
#> 21      9.19 0.234 72      8.72      9.65
#> 22      7.59 0.234 72      7.13      8.06
#> 23      5.73 0.234 72      5.26      6.19
#> 24      5.05 0.234 72      4.58      5.51
#>

```

```

#> Confidence level used: 0.95

```

```

model_means_cld <- multcomp::cld(q2_emm, adjust = "Tukey", Letters = letters, alpha = 0.05)
model_means_cld

```

```

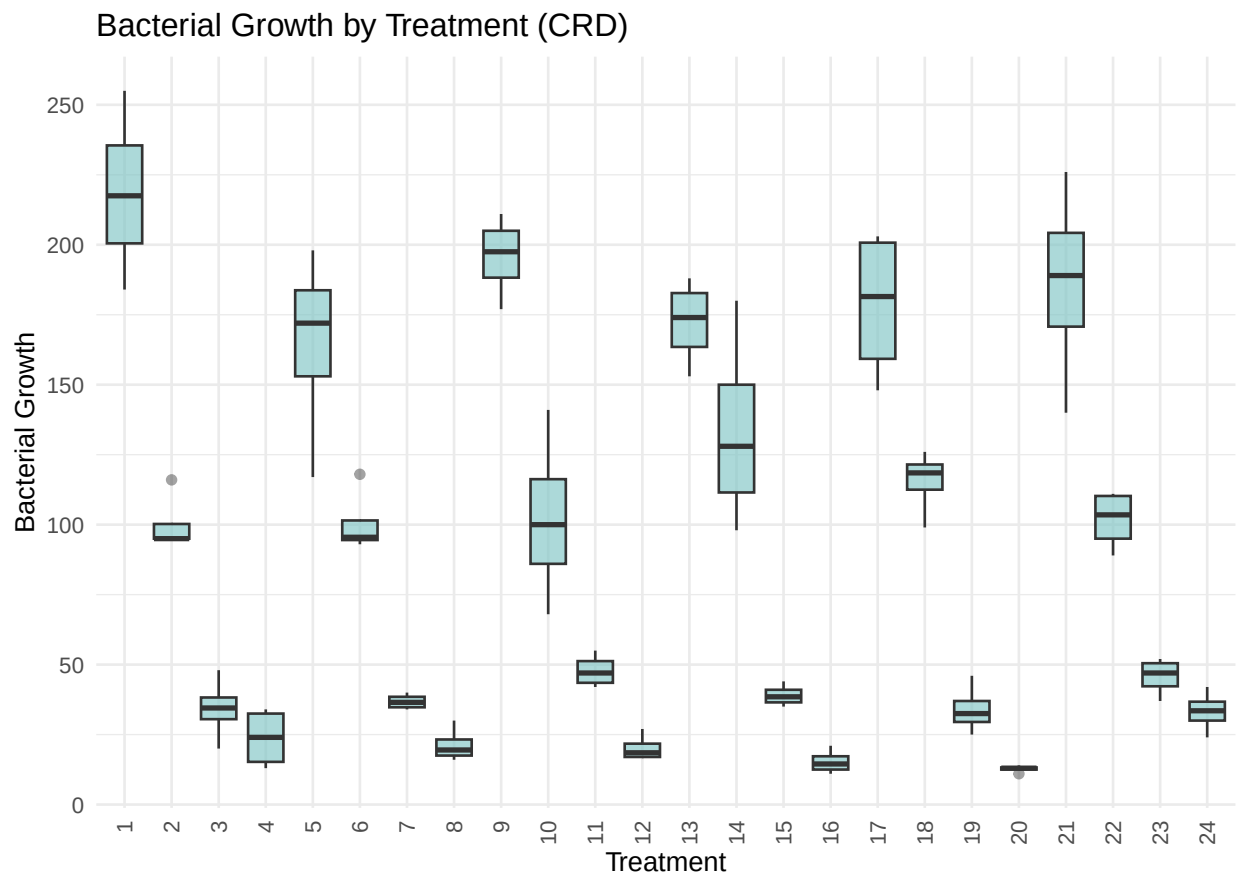
#>  treat emmean      SE df lower.CL upper.CL .group
#> 20      3.31 0.234 72      2.57      4.06      a
#> 16      3.58 0.234 72      2.84      4.33      a
#> 12      4.10 0.234 72      3.35      4.84     ab
#>  8      4.17 0.234 72      3.42      4.92     abc
#>  4      4.29 0.234 72      3.55      5.04     abc
#> 24      5.05 0.234 72      4.30      5.79     bcd
#>  3      5.06 0.234 72      4.32      5.81     bcd
#> 19      5.08 0.234 72      4.34      5.83     bcd
#>  7      5.28 0.234 72      4.53      6.02     bcd
#> 15      5.40 0.234 72      4.65      6.14      cd
#> 23      5.73 0.234 72      4.98      6.47      d
#> 11      5.82 0.234 72      5.08      6.57      d
#> 10      7.55 0.234 72      6.80      8.29      e
#>  2      7.56 0.234 72      6.81      8.30      e
#>  6      7.56 0.234 72      6.82      8.31      e
#> 22      7.59 0.234 72      6.85      8.34     ef
#> 18      7.92 0.234 72      7.17      8.66     efg
#> 14      8.25 0.234 72      7.51      9.00     efgh
#>  5      8.84 0.234 72      8.09      9.58     fghi
#> 13      9.00 0.234 72      8.25      9.74      ghi
#> 17      9.08 0.234 72      8.34      9.83      ghi
#> 21      9.19 0.234 72      8.44      9.93      hi
#>  9      9.36 0.234 72      8.61     10.11      hi
#>  1      9.67 0.234 72      8.92     10.41      i
#>

```

```
#> Confidence level used: 0.95
#> Conf-level adjustment: sidak method for 24 estimates
#> P value adjustment: tukey method for comparing a family of 24 estimates
#> significance level used: alpha = 0.05
#> NOTE: If two or more means share the same grouping symbol,
#>       then we cannot show them to be different.
#>       But we also did not show them to be the same.
```

Figure

```
ggplot(q2_data, aes(x = treat, y = growth)) +
  geom_boxplot(fill = "#80c5c5", alpha = 0.65, outlier.alpha = 0.45) +
  labs(
    title = "Bacterial Growth by Treatment (CRD)",
    x = "Treatment",
    y = "Bacterial Growth"
  ) +
  theme_minimal(base_size = 11) +
  theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust = 1))
```



Final conclusion for Question 2:

Raw assumptions failed, Box-Cox transformation corrected assumptions, ANOVA confirmed strong treatment differences, and treatment 20 showed the lowest bacterial growth.